Streamlining Academic Leave Processes: A Web-Based Leave Management System for Educational Institutions

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Abstract— Leave management is an important administration practice in learning institutions. The traditional processing of the leave requests in most institutions is through manual means using paper forms or informal communication and is thus likely to be delayed, prone to errors and lack of good records. This paper will address these problems by offering a web-based Leave Management System, which will facilitate in automating this process of leaves application, approval and monitoring. It allows the students and members of staff to request by means of an online interface and the requests are reviewed, approved or declined by the authorities such as the Head of Department and the administrators with special dashboards. The system is developed using the most recent web technologies including React and the frontend, Node.js and Express as the backend services and MongoDB to store the database. It has core capabilities, such as JWT-secured log-in, role-based access control, automated email notifications via Nodemailer, live in-app notifications via a bell icon, automatic update of attendance after leave approval and an inbuilt analytics dashboard. The system enhances transparency and minimizes paper work in addition to offering a feasible system of keeping the leave records in the education institutions by creating a digitalization of the leave management procedure.

Keywords—Leave Management System, Web-Based Application, JWT Authentication, Educational Institutions, Automation, React, Node.js, MongoDB.

I. INTRODUCTION

The management of leave is a very vital administrative procedure in schools since the students and the faculty staff tend to require the leave either academically, personally, or even due to medical reasons. Most of the institutions are still managing leave requests manually through the use of paper forms or informally. This conventional method may result in processes that are slow to approve and lack transparency and inability to keep proper records. The higher the number of users, the more time consuming and ineffective to the administrators to handle leave application manually. As the web technologies improved, most of the administrative processes can be automated so as to enhance efficiency and availability.

A web-based Leave Management System is a digital platform on which the user can enter the leave request and monitor the status in real time. The requests can be viewed and managed by the authorities, including Head of Department and administrators, via special dashboards. The system suggested will decrease the amount of paper work, enhance communications and guarantee good record keeping in the educational institutions, by automating the leave management process in the organization.

II. LITERATURE SURVEY

The latest works devoted to the systems of managing the field of administration have shown the relevance of the digital platform in terms of enhancing the efficiency of an organization. The conventional ways of managing leaves in institutions of learning are based on manual paper work and physical approval of leaves that may cause delays, inconsistencies in data and a challenge in keeping proper records. Scientists have used web-based management systems to automate these administration activities, and make them more accessible to both the users and administrators.

Certain researches have suggested online leave management systems whereby employees or students can make their leave requests using a central platform that is web-based. Such systems allow authorities to screen and give approvals to requests effectively and have records that are stored in organized databases. Nevertheless, there are still a number of available systems which have issues regarding the scope of scaling, usability and efficient role-based access control. Recent advances in the web technologies have made it possible to develop more valid and convenient platforms that allow to track real-time, store data safely and communicate effectively between users and administrators. These innovations promote designing an improved web-based system of managing leaves that may simplify the process of

managing leaves and enhance the amount of transparency in colleges.

A. Problem Statement

1) Research Question: What is the best way to enhance efficiency of leave application, approval, and record management in the institutions of learning through use of web-based Leave Management System that is transparent and can handle data securely?

2) Relevance:

a. Administrative Inefficiency: Most institutions have manual process of taking leave and this requires use of physical copies and multiple approvals thus making the workload of the administrative staff heavy and creates delays.
b. Record Management Issues: Maintaining paper documents of leave makes it difficult to trace leave history, retrieve previous details as well as ensure accuracy in data.
c. Having Transparency and Accessibility: Both students and the members of the faculty will find it hard to track the status of their respective leaves.

III. PROPOSED SYSTEM

The proposed solution is a web-based Leave Management System that aims at automating the leave application and approval process and monitoring of the same in educational institutions. The system will substitute the former manual process of leaves processing by integrating a centralized digitized system through which students and faculty members will be able to post leave requests through the online system. The requests are then sent to their respective authorities, including Head of Department (HOD) or administrator to be verified and authorized.

This system is powered with the latest web technologies, in which the frontend interface is written with React to deliver a responsive and interactive interface to the user. The server-side operations and business logic are provided with the help of Node.js and Express. The database is MongoDB which stores the details of the users, leave applications, and approval records. The system also enables uploading of file leaves proof and role-based dashboards to the various users.

The suggested system enhances efficiency as less paperwork is done manually, leaving follow-up of leave real-time, and keeping the digital records organized. It is also very useful in improving communication between the users and the authorities via the system dashboard that gives the user a status update of the leave application.

The proposed Leave Management System has a number of modules managing various capabilities of the system. A particular module has a set of tasks to carry out to facilitate orderly running of the leave management process.

A. System Modules

A. Student Module

This module enables students to access the system and place leave requests by filling the required information that includes the length of the leave, the cause of the leave, and documents to support the leave request. Through their dashboard, students are also able to monitor their leave application status and their attendance.

B. Faculty Module

Faculty module enables faculty members to preview the requests of students who leave that have been allocated to them. The request can be verified by the faculty members and sent to the Head of Department who can approve it further. Mark daily attendance for department students.

C. HOD Module: Allows the Head of Department to review and approve or reject leave requests at the second approval level for both students and faculty, view department-wise leave reports, and receive notifications for pending approvals.

D. Principal Module

Provides the Principal with final approval authority over all leave requests across all departments, access to institution-wide reports and analytics, and real-time notification of pending approvals.

E. Admin Module

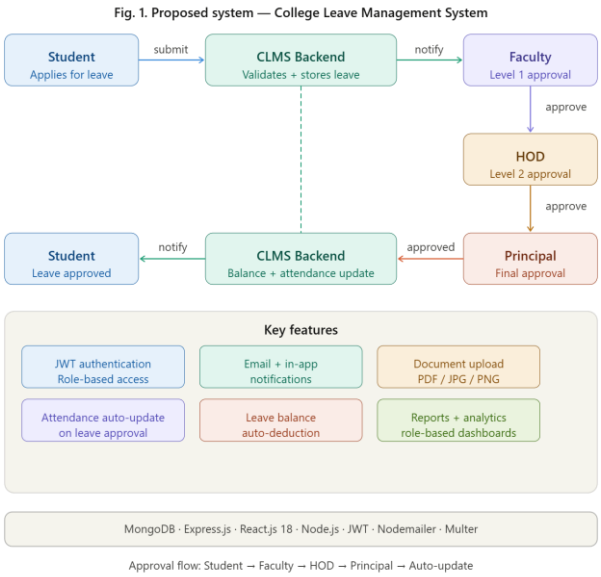
The duty of the admin module is to manage the system in general. The administrator is able to administer user accounts, keep track of leave history and maintain database. The administration also makes sure that the system is running in a smooth and safe manner.

F. Approval Process

The approvals leave module happens to be the one that is charged with the responsibility of approving the submitted leave requests. The authorized persons like the faculty advisors and the Head of the Department do the facility of recognizing the leave requests that the applicants have typed in this module. The request is verified by the faculty advisor who either approves or disapproves the request. In case of approval of the request, it is sent to the Head of the Department to be approved. The system will change the status of the leave respectively and inform an applicant of the decision. This module makes the institution a temporality and ensures that approval is done systematically in the institution.

B. Multi-Level Approval Process

The approval chain is dynamically determined based on the applicant's role. For a student, the chain is Faculty → HOD → Principal. For a faculty member, the chain is HOD → Principal. For an HOD, only the Principal's approval is required. The approval chain is stored as an embedded array within the leave document in MongoDB. At each stage, automated email notifications are sent using Nodemailer and in-app notifications are created in the Notifications collection. Upon final approval, leave balances are automatically deducted and attendance records are updated.



IV. METHODOLOGY

The development of the system was done in an iterative agile approach with clear stages of requirements analysis, system design, implementation, testing and installation.

A. Technology Stack

Component	Technology
Frontend	React.js 18, React Router v6, Axios
Backend	Node.js, Express.js
Database	MongoDB 8.0, Mongoose ODM
Auth	JWT, bcryptjs
Email	Nodemailer, Gmail SMTP
Upload	Multer (PDF, JPG, PNG)

Table I. Technology Stack

B. Database Design

The MongoDB database consists of five collections that comprise Users, Departments, Leaves, Attendance and Notifications. Users collection stores the role, credentials, department reference and leave balance. The Leaves collection has the approval chain in it which is a set of sub-documents having approver reference, role, status, comment and action date. The built-in design does not require complex joins, and it makes it easier to track the workflow states.

C. Authentication Flow

Upon user login, the backend authenticates the information given by the user using the compare function of bcrypt and provides a signed JWT token with an expiration duration of 7 days. The frontend saves the token in localStorage and adds it to every API request through an Axios interceptor. Exists Before a request is served, middleware Backend route protection validates the token and applies role-based access control.

D. Implementation

Its backend is modular in structure and includes separate authentication, leave management, user management, attendance tracking, department management, notifications and reports controllers. It is written on the front end in functional components with hooks, in React, having an AuthContext global state and custom CSS to make it look good, without depending on a third-party framework.

V. RESULTS

College Leave Management System was installed and tested with all five user roles with seeded demonstration data. The system has managed to showcase the entire leave process of application to the ultimate approval. The multi-level approval workflow was working as expected in the test situations. The leave requests that were presented by the students were automatically forwarded to the faculty advisor to have the initial approval. On approval, requests went to the HOD and then to the Principal. Email messages were sent on a user-stage basis and notification badges in the app were updated within 30 seconds through the polling system. The upload of documents worked well with PDF, JPG, and PNG files of 5MB size. Final approval and attendance records were properly deducted and changed automatically without any human intervention. Role-based dashboards were used to show the correct real-time statistics and the admin panel was used to manage users and departments fully.

Feature	Result
Approval Workflow	Pass ✓
Email Notification	Pass ✓
In-App Notification	Pass ✓
Document Upload	Pass ✓
Attendance Update	Pass ✓
Role-Based Access	Pass ✓
Leave Balance Deduction	Pass ✓
Admin User Management	Pass ✓

Table II. Test Results Summary

The system showed that it had great benefits over the manual system of leave management such as removing paperwork, real time status availability, automatic record keeping and better communication among the all the stakeholders as a result of automation notification.

VI. CONCLUSION

The paper involved the design and implementation of a web based College Leave Management System in the MERN stack. The system has been able to automate the process of leave application and approval by multiple levels in educational institutions, with five user roles, role-based dashboards, and access control.

The major contributions of this system are dynamic multi-level approval process, automatic email and in-app notification, support of leave evidence by uploading

documents, automatic integration with attendance and the ability to report comprehensively. The system was well tested and all significant functional requirements met. The MERN stack turned out to be a helpful technology platform that offered a complete JavaScript development platform, versatile data modeling with MongoDB, and a user interface that is responsive with React.js.

The future developments will involve creating a mobile application, using cloud storage to manage documents, real-time notification using the WebSocket protocol, integration of biometrics attendance, and deployment on a cloud platform to make the system available to every part of the institution.

VII. FUTURE SCOPE

- *Mobile Application:* Create a mobile application to ensure people can apply and manage their leave comfortably using the mobile phones.
- *Notification System:* Provide SMS notifications to alert users on a real-time basis on either approving or rejecting leave.
- *Leave Analytics:* Introduce the use of graphical reports and analytics to determine leave trends and patterns.
- *Attendance Integration:* Combine the system with attendance or biometric systems to make sure of accurate leaves. Cloud deployment: The system should be deployed on cloud platforms in order to have an improved scalability, security and accessibility.
- *AI Chatbot Support:* Add AI chatbot to assist the users with leave application and frequently asked questions.
- *Cloud Storage Integration*: Swap a on-premise file storage with AWS S3 or Cloudinary to have the reliable and scalable management of documents.
- Push notifications: Use WebSocket or Firebase Cloud Messaging as a replacement to the polling mechanism to generate instant notifications.
- Biometric Attendance Integration: Add biometric devices.
- Certificate Creation: Issue official certificates in PDF format to students that attend events in an automatic manner.

VIII. REFERENCES

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